THE 15-PUZZLE: BETTER HEURISTICS, LESS SEARCH

EC3161 – Artificial Intelligence | Mini Project P02

Submitted by: KOLLIPARA SAI SPOORTHI
Roll Number: 624132
Department of Electronics and Communication Engineering
National Institute of Technology Andhra Pradesh

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1. Overview

The 15-puzzle is a 4 × 4 sliding-tile puzzle containing fifteen numbered tiles and one blank space. This project compares three heuristic approaches using Iterative Deepening A* (IDA*): Manhattan Distance, Manhattan Distance with Linear Conflict, and an additive Pattern Database.

Main result: The PDB heuristic reduced node expansions by 79.7% compared with Manhattan Distance on the second test instance.

2. Problem Formulation

- **State:** Arrangement of fifteen tiles and one blank space.
- **Initial state:** A scrambled puzzle configuration.
- **Goal state:** Tiles arranged in numerical order, with the blank at the bottom-right.
- **Actions:** Move the blank up, down, left, or right.
- **Path cost:** 1 per move.
- **Objective:** Find the minimum-move solution.

3. Methods

3.1 Baseline – Manhattan Distance

Manhattan Distance calculates the sum of horizontal and vertical distances of all tiles from their goal positions.

[
h_{MD}=\sum_{i=1}^{15}
(|r_i-r_i^|+|*c_i-c_i*^|)
]

3.2 Improved Heuristic – Linear Conflict

Linear Conflict adds 2 for each disjoint pair of conflicting tiles to Manhattan Distance.

[
h_{LC}=MD+2LC
]

3.3 Improvement – Additive Pattern Database

The fifteen tiles are divided into five disjoint patterns:

Pattern	Tiles
P1	1, 2, 3
P2	4, 5, 6
P3	7, 8, 9
P4	10, 11, 12
P5	13, 14, 15

Each database is constructed using backward 0–1 BFS. Moves of pattern tiles have cost 1, while moves of non-pattern tiles have cost 0.

The final heuristic is:

[
h_{PDB}=\sum_{i=1}^{5}PDB_i
]

Each pattern contains 43,680 states, giving 218,400 entries in total.

4. Experimental Results

Environment: Windows 11, Python 3.13.14, Python standard library.

PDB construction time: **0.47 seconds**.

Three instances were generated using fixed scramble lengths and random seeds.

Test	Scramble	Seed
1	20	10
2	30	20
3	40	30

*Table 1. IDA performance comparison**

Test	Heuristic	Moves	Expanded Nodes	Time (s)
1	MD	20	31	0.0004
1	MD+LC	20	31	0.0037
1	PDB	20	30	0.0006
2	MD	30	16,550	0.1626
2	MD+LC	30	9,527	0.4765
2	PDB	30	3,365	0.0506
3	MD	26	79	0.0008
3	MD+LC	26	45	0.0032
3	PDB	26	26	0.0007

The PDB heuristic expanded the fewest nodes in all three test instances.

For Test 2, node reductions were:

- **79.7%** compared with Manhattan Distance.
- **64.7%** compared with Linear Conflict.

Linear Conflict required more execution time than Manhattan Distance in Test 2 despite expanding fewer nodes, indicating additional heuristic evaluation overhead.

5. Analysis

The experimental results show that:

- Manhattan Distance provides a simple baseline.
- Linear Conflict reduces node expansions by providing additional heuristic information.
- PDB achieves the lowest node expansions in all three test instances.
- Stronger heuristics can reduce search effort, but their evaluation may introduce computational overhead.

All three heuristics returned the same solution lengths for the tested instances.

6. Correctness Evidence and Limitations

The implementation uses admissible heuristic approaches and additive cost partitioning for PDB. All three methods produced identical solution lengths on the tested instances.

However, independent verification using a trusted exact solver and explicit solution-path replay validation were not performed.

Other limitations include:

- Only three final test instances were evaluated.
- A* comparison was not implemented.
- Memory usage and effective branching factor were not measured.

7. Reproducibility

Software environment:

- Operating System: Windows 11
- Python: 3.13.14
- External libraries: None

Run command:

```
None
python puzzle15.py
```

PDB construction is performed before the search experiments. Search execution times should be interpreted separately from the reported PDB construction time.

Hardware: Processor:Intel(R) Core(TM) i7-8665U CPU @ 1.90GHz (2.11 GHz)
Installed RAM:16.0 GB (15.8 GB usable)

8. AI-Use Statement

AI assistance was used for understanding concepts, debugging support, code explanations, and improving project documentation. Implementation and experiments were carried out using the project code.

9. References

1. EC3161 – Artificial Intelligence, Mini Project Assignment, NIT Andhra Pradesh.
2. Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, 4th Edition.
3. Richard E. Korf, “Depth-First Iterative-Deepening: An Optimal Admissible Tree Search,” 1985.
4. Richard E. Korf and Ariel Felner, “Disjoint Pattern Database Heuristics,” 2002.